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Challenges, developments and perspectives in intermittent river ecology

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SUMMARY

1. Although more than half the world's river networks comprise channels that periodically cease to flow and dry [intermittent rivers (IRs)], river ecology was largely developed from and for perennial systems. Ecological knowledge of IRs is rapidly increasing, so there is a need to synthesise this knowledge and deepen ecological understanding.

2. In this Special Issue, we bring together 13 papers spanning observational case studies, field and laboratory experiments and reviews to guide research and management in this productive field of freshwater science. We summarise new developments in IR ecology, identify research gaps and needs, and address how the study of IRs as highly dynamic ecosystems informs ecological understanding more broadly.

3. This series of articles reveals that contemporary IR ecology is a multifaceted and maturing field of research at the interface between aquatic and terrestrial ecology. This research contributes to fresh water and general ecology by testing concepts across a range of topics, including disturbance ecology, metacommunity ecology and coupled aquatic-terrestrial ecosystems.

4. Drying affects flow continuity through time and flow connectivity across longitudinal, lateral and vertical dimensions of space, which aligns well with the recent emphasis of mainstream ecology on meta-system perspectives. Although most articles here focus on the wet phase, there is growing interest in dry phases, and in the terrestrial vegetation and invertebrate assemblages living in and along IR channels. We encourage interdisciplinary studies on IRs to further blur the demarcation between aquatic and terrestrial ecosystems and develop more integrated perspectives.

5. As a result of climate change and human modification of landscapes and waterways, flooding and drought are expected to become more extreme and widespread. Shifts in streamflow regimes from perennial to intermittent may exacerbate the duration and frequency of dry phases in IRs with serious implications for river ecosystems and the quality and diversity of services they provide.

Keywords: drying, flow cessation, ephemeral, intermittent, temporary, rivers, climate change, genealogy

Introduction

River ecology initially developed from and for the study of perennial rivers and streams (Datry, Larned & Tockner, 2014a). Yet more than half the channels comprising river networks globally cease to flow or dry periodically (Acuña *et al.*, 2014; Datry *et al.*, 2014a), with the

proportion exceeding 80% in some regions (Levick *et al.*, 2008; Sabater & Tockner, 2010). Moreover, the frequency and duration of flow cessation and drying are increasing in many rivers in response to climate change and growing demands for fresh water (McKerchar & Schmidt, 2007; Larned *et al.*, 2011; Jaeger, Olden & Pelland, 2014). Several large and once-perennial rivers no longer flow

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continuously to their deltas, including the Yellow (China), Colorado (U.S.A), Amu and Syr Darya (Central Asia) and Nile (Egypt) (Gleick, 2003; Postel & Richter, 2012), and the number of perennial systems becoming intermittent is expected to increase (Larned *et al.*, 2010; Döll & Schmied, 2012; Jaeger *et al.*, 2014). Rivers and streams without continuous flow have been named by a variety of terms, such as non-perennial, temporary, intermittent, ephemeral, seasonal and episodic (Uys & O'Keeffe, 1997; Williams, 1996; Arthington, Bernardo & Ilhéu, 2014). For the sake of consistency, generality and simplicity, we refer to them here as intermittent rivers (IRs): any river or stream that ceases to flow or dries at any point in time and space (Acuña *et al.*, 2014; Datry *et al.*, 2014a), whether naturally or as a result of human activities.

After years of relative obscurity, the ecological study of IRs is trending upward. The interest from river scientists and managers in IRs has grown rapidly in the past decade (Datry, Arscott & Sabater, 2011; Arthington *et al.*, 2014) and the number of IR studies published in freshwater and ecology journals, including *Freshwater Biology*, has grown considerably (Fig. 1). According to the results of a Web of Science search conducted by Leigh *et al.* (2016a) in 2014, over 70% of IR literature published in these journals on the topics of fish and invertebrate ecology, biogeochemistry, and hydrological and ecological-condition assessment has appeared since the mid-2000s (Fig. 1). Ecological research on IRs now appears in high-rank (e.g. Raymond *et al.*, 2013; Acuña *et al.*, 2014; Jaeger *et al.*, 2014) and general ecology journals (Boersma *et al.*, 2014; Datry *et al.*, 2014a,b; Cañedo-Argüelles *et al.*, 2015). Since 2008, almost every major freshwater conference has had a special session on IRs in its programme, producing special issues of journals on the topic of IRs or related themes such as hydrological (dis)connectivity (e.g. Nadeau & Rains, 2007; Datry *et al.*, 2011; Arthington *et al.*, 2014). Several international projects on IRs have been initiated, including the MIRAGE European project (www.igb-berlin.de/mirage.html, 2009–2011) and its continuation as a Life project (www.lifetrivers.eu, 2014–2016), the international Intermittent River Biodiversity Analysis and Synthesis program (www.irbas.fr, 2013–2016), the 1000 Intermittent River Project (http://1000_intermittent_rivers_project.irstea.fr, 2015–) and a H2020 European COST Action (Science and Management of Intermittent Rivers and ephemeral streams (http://www.cost.eu/COST_Actions/ca/CA15113, 2016–2020). This growing interest in IRs from scientists and managers indicates the prevalence and specificities of IRs, but also the scientific contribution that the study

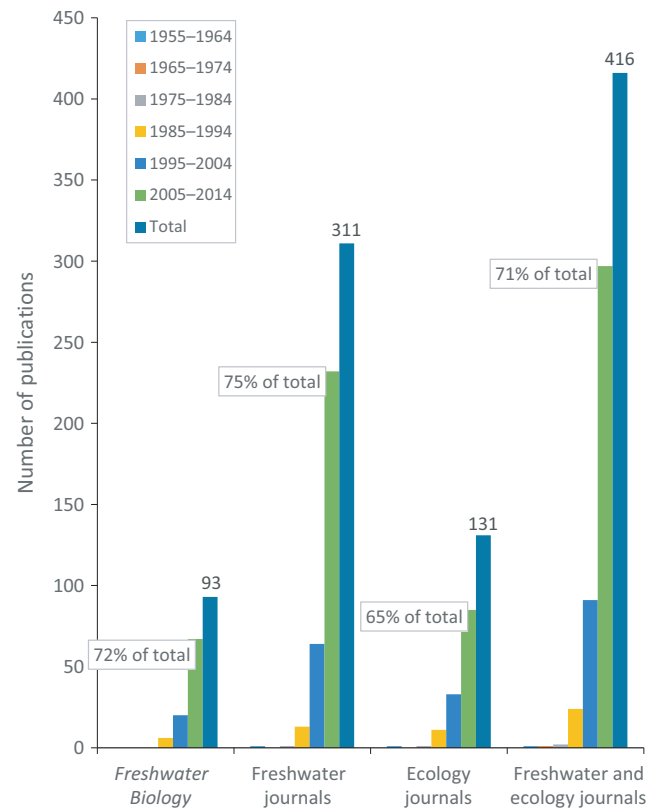


Fig. 1 Number of IR-related publications on the combined topics of fish ecology, invertebrate ecology, biogeochemistry, hydrological assessment and ecological-condition assessment appearing in freshwater and ecology journals, and *Freshwater Biology* specifically, through time. Freshwater journals include any journal with the character string *water or aquatic in the title, ecology journals include the string *ecol in their titles, and freshwater and ecology journals include one or more of these three strings in their titles. Data taken from a Web of Science literature search conducted by Leigh *et al.* (2016a).

of IRs can make to general ecology and the growing awareness of the social, cultural, economic and environmental values of these systems.

From humble beginnings, the future of IR ecology is bright

In this Special Issue, we specifically seek to examine how IR ecology is maturing and how some key questions in ecology can be addressed through the study of IRs as highly dynamic ecosystems. We bring together 13 papers, including this introduction, spanning observational case studies (Corti & Datry, 2016; Datry, Bonada & Heino, 2016a; Datry *et al.*, 2016b; Marshall *et al.*, 2016; Rolls, Heino & Chessman, 2016; Siebers *et al.*, 2016; Stromberg & Merritt, 2016; Welter & Fisher, 2016; Whitney *et al.*, 2016), field (Vander Vorste, Malard & Datry,

2016) and laboratory (Stubbington *et al.*, 2016) experiments, and reviews (Leigh *et al.*, 2016a).

Leigh *et al.* (2016a) review the basic and applied ecological literature on IRs, document the genealogy of ideas and highlight major advances across several research themes. While IR studies were instrumental in the development of fluvial geomorphology theory (e.g. Leopold & Miller, 1956; Schumm & Hadley, 1957), research on temporary standing waters (e.g. Wiggins, Mackay & Smith, 1980) and perennial systems (e.g. Harrel, Davis & Dorris, 1967) was influencing early work on IRs. Persistent research themes that have informed ecological theory and IR management include resistance and resilience to drying. Leigh *et al.* (2016a) show through their review that the recent boom in IR publications coincides with the increasing need for science to inform policy concerning the legal status of IRs, and recognition that the number and length of IRs are likely to increase with climate change and growing human demands for fresh water. In aggregate, IR research has a strong future with growing interest around the world and across multiple research themes.

IRs are highly dynamic ecological systems: landscape mosaics, biogeochemistry and biological organisation in IRs

Landscapes change over time through physical and biological drivers that are patchily distributed, spatially and temporally (Bormann & Likens, 1979). Although terrestrial landscapes and branching riverine networks differ in many respects, the terrestrially based concept of shifting habitat mosaics applies well to rivers because of their dynamic nature, the interrelationships between hydrology, fluvial geomorphology and the distribution of biota, and water as a vector linking patches (Ward, Malard & Tockner, 2002; Stanford, Lorang & Hauer, 2005; Altermatt, 2013). Datry *et al.* (2016a) use hydrogeographic data collected by citizen scientists to describe the spatiotemporal dynamics of IR habitats within five river networks in west central France, their hydrologic controls, and associated extinction risk and colonisation for hypothetical fish and aquatic insects. However, general configuration and compositional patterns of IR habitats were found among the networks studied, a high degree of complexity was seen in landscape-metric variability over relatively short time frames and among neighbouring catchments. Describing IRs with landscape metrics adds spatial dimensions that are typically missed by flow metrics alone and may be particularly applicable to internally coupled aquatic-terrestrial

systems (i.e. dry, lentic and lotic conditions within and across reaches). Datry *et al.* (2016a) propose that a landscape approach not only advances ecological understanding within IRs, but that this perspective will further advance IRs as models for studying network dynamics in general.

The spatiotemporal variability in flow and mosaic composition of IR landscapes (dry, lentic and lotic patches) poses particular challenges for determining the extent of material transport, transformation and retention across networks (Malard, Tockner & Ward, 1999; Larned *et al.*, 2010; Sponseller, Heffernan & Fisher, 2013). Transport of nutrients and organic matter from terrestrial hill slopes into intermittent channels can generate hot spots of microbial activity, but this may depend on the spatial extent of the hydrologic flow paths generated by rainfall pulses and how long resources, including water, remain available. Welter & Fisher (2016) show how storm characteristics influence the extent of hydrologic connectivity along a terrestrial-aquatic flowpath in the Sycamore Creek catchment of Arizona, U.S.A., and thereby affect denitrification and nitrogen retention. They show that the size, timing and intensity of storms control hydrologic connectivity in small streams and that riparian terrace and hillslope soils have the highest denitrification potential. Deep and moist channel sediments (i.e. the hyporheic zone) have low denitrification potential but contribute to nitrogen loss for a longer period of time overall. The study reveals important links between rainfall pulses, vertical and horizontal hydrologic connectivity and the magnitude of transport, storage and processing of nutrient and organic material in arid IR landscapes.

Intermittent rivers in many regions are renowned for their high levels of flow variability (e.g. Puckridge *et al.*, 1998; Poff *et al.*, 2006; Bonada *et al.*, 2007), which can profoundly affect biogeochemical processes, creating 'hot spots' and 'hot moments' within IRs (Fisher *et al.*, 2001; Datry *et al.*, 2014a). As surface flows become discontinuous and pools contract, particulate and dissolved organic matter (DOM) become concentrated and can subsequently change in composition through the drying phase. Siebers *et al.* (2016) investigate the flow-related causes of such changes in pools of dryland Australian IRs using a combination of $\delta^{18}\text{O}$ and $\delta^2\text{H}$ values of surface and alluvial ground waters and DOM fluorescence excitation–emission spectroscopy. Although the extent to which pools remain connected to alluvial ground water is strongly linked to changes in fluorescence, the findings suggest that high variability in groundwater-surface water mixing contributes to substantial variability in

compositional patterns among pools. This study emphasises that variation in biogeochemical processes and DOM characteristics within these systems cannot be attributed to the presence or absence of surface flow alone, highlighting the complexity of IR flow regimes and hydrological-(dis)connectivity dynamics.

Landscape characteristics and hydrological-(dis)connectivity dynamics can also interact with species traits to influence local extinction, connectivity and movement behaviour of biota (Fausch *et al.*, 2002; Datry *et al.*, 2016b). In many dryland IRs, for example, fish can be confined to isolated waterholes for much of the year and are able to colonise the network only during brief flow events. The ways fish use waterholes over time and space to either re-establish populations following dry periods (resilience) or maintain populations during dry periods (resistance) are key determinants of population persistence (Jaeger *et al.*, 2014). Marshall *et al.* (2016) tagged 215 individuals of three common species of large-bodied fish and monitored their movements in and between waterholes of the Moonie River (Queensland, Australia) for 3 years. Although some individuals of all species moved during flow events without clear upstream or downstream preference, others remained within the same waterhole throughout the study. Timing of flow resumption, independent of its magnitude or duration, was a key trigger for fish movement, and many individuals returned to the same waterhole after flow ceased. Maintaining such behaviour in a highly variable environment may improve survival and management and conservation strategies should promote movement opportunities for fish at large spatial scales to preserve population resilience.

Predicting responses of terrestrial vegetation in the riparian zones of IRs (and other river types) to changes in hydrological (dis)connectivity and water availability is also vital for effective river management and conservation. Terrestrial plant diversity, composition and health not only are affected strongly by water availability but also can influence water availability via their effects on evapotranspiration and the terrestrial water cycle (Stromberg *et al.*, 2007; Sterling, Ducharme & Polcher, 2013; Van Dijk *et al.*, 2013). Stromberg & Merritt (2016) use functional traits that capture the mechanisms underlying habitat preferences and tolerances of dryland riparian plants to classify guilds associated with ephemeral, intermittent and perennial reaches of Arizonan rivers. The use of functional traits presents a transferable approach that goes beyond taxonomy and simple life form or region-of-origin classification, and can be used to predict changes in plant guild and trait composition

in response to shifts in flow intermittence and flood frequency. In particular, their findings emphasise the need to conserve ephemeral and intermittent reaches, not just the rare perennial reaches, within river networks of arid landscapes to maximise plant-guild diversity.

Revival of disturbance ecology: community resistance and resilience in IRs at the aquatic-terrestrial interface

Many mechanisms contribute to the well-known resilience of aquatic invertebrate communities in IRs to periodic drying (Stanley *et al.*, 1994; Fritz & Dodds, 2004; Vander Vorste *et al.*, 2015; Leigh *et al.*, 2016a). Challenging the view that drift is the primary mechanism, Vander Vorste *et al.* (2016) suggest that other processes such as colonisation from the hyporheic zone may be equally important. By manipulating both flow and drift, and creating artificial dry spells in channels of a braided alluvial IR, they show that community resilience is not significantly altered by the absence of drift. The results instead point to the hyporheic zone as the main source of colonists. The importance of the hyporheic zone as a vertical refuge from drying in IRs depends on multiple factors, including physical habitat characteristics (e.g. sediment porosity) and biotic features (e.g. invertebrate behaviour and morphology; Stubbington, 2012). Vander Vorste *et al.* (2016) emphasise the need to consider all potential sources of colonists as contributors to community resilience across the entire spectrum of IRs and invertebrate inhabitants.

Resistance as well as resilience is important in promoting biological persistence in IRs. Stubbington *et al.* (2016) investigate one such resistance mechanism by examining and rehydrating the 'seedbank' of IRs, which comprises the life stages of aquatic invertebrates that remain viable in bed sediments during dry phases. Their results provide some support for the hypothesis that longer dry phases decrease the abundance and richness of the seed bank but also highlight the complexity of wet-dry cycling, seasonality and invertebrate life-cycle relationships. Their study adds to the scant knowledge of IR seedbank ecology (e.g. Storey & Quinn, 2013; Stubbington & Datry, 2013) and, we hope, will inspire researchers to devise experiments to disentangle effects of environmental variables that influence biotic responses to drying and survival in dry riverbeds.

Indeed, evidence indicates that dry riverbeds are not ecologically dead (e.g. Steward *et al.*, 2012; Stubbington *et al.*, 2016; Vander Vorste *et al.*, 2016) and there are likely complex interactions between dry/flowing (or

terrestrial/aquatic) riverbed communities and ecosystem processes awaiting explanation. However, while there has historically been interest in studying community patterns at terrestrial–aquatic interfaces (e.g. Reich, 1991; O’Callaghan *et al.*, 2013; Langhans & Tockner, 2014), characterisation of community dynamics involving complete terrestrial–aquatic shifts, as seen in IRs, has received considerably less attention (but see Moon, 1956; Plum, 2005; Steward *et al.*, 2012). Corti & Datry (2016) describe terrestrial and aquatic invertebrate communities from the active channel and the riparian zone of intermittent and perennial reaches along a French river. Aquatic and terrestrial taxa at intermittent reaches (during flowing and dry conditions respectively) were largely subsets of the taxa at perennial reaches and the adjacent riparian zone, respectively, indicating strong longitudinal and lateral biological connections to upstream perennial reaches and bank habitats respectively. Among-reach variation in community composition contributed most to overall terrestrial diversity, whereas within-reach variation was the greatest contributor to overall aquatic diversity. These findings deny the assumption that the dry phase of IRs is a lifeless period and emphasise that IRs are coupled terrestrial–aquatic ecosystems.

IRs as models to test and develop metacommunity concepts

Following current developments in community ecology and biodiversity research, metacommunity perspectives are increasingly being applied in IR studies (e.g. Datry *et al.*, 2016a). Patterns of nestedness and species turnover can be due to two non-mutually exclusive processes. Common species of high dispersal ability can colonise more patches in the network, leading to large population sizes and ranges (Albanese, Angermeier & Peterson, 2009), than can species with limited dispersal ability. Common species also tolerate harsh environmental conditions and persist throughout a wide range of habitats, resulting in low extinction rates and large population sizes and ranges (Schlosser, 1990; Taylor & Warren, 2001). Although these processes were integrated in a metacommunity model by Larned *et al.* (2010), stating that longitudinal patterns of nestedness across connected stream habitats are caused by dispersal limitation, whereas temporal patterns of nestedness within habitats during stream drying are caused by interspecies differences in environmental tolerance, there is still little empirical evidence for this model (but see Datry *et al.*, 2014b). Whitney *et al.* (2016) quantify species abundance

distributions and the importance of nestedness and turnover to community dissimilarity in tallgrass prairie streams of eastern Kansas, U.S.A. Few species were common but nestedness drove community dissimilarity across sites, notably during the first phases of recolonisation. Although common species were the first to colonise and the last to go extinct locally, the study raises the question: is commonness predictive of rather than predicted by colonisation and extinction?

Datry *et al.* (2016b) further examine the relative roles of dispersal and environmental filtering in structuring biological communities in dynamic landscapes. When one thinks of IRs, tributaries of the Amazon River typically do not come to mind. Datry *et al.* (2016b) compare the local and regional diversity of aquatic invertebrate and fish communities from intermittent and perennial reaches of rivers in central Bolivia to see if natural drying has a similar effect on the structure and composition of Neotropical communities as has been observed in other regions. Intermittent and perennial reaches had similar invertebrate assemblages but perennial reaches had higher fish densities and richness than intermittent reaches. Using metacommunity analyses, in combination with life history knowledge of the local fauna, the authors were able to infer the relative importance of dispersal compared to local environmental filtering in structuring aquatic invertebrate and fish communities across the study sites. The study further suggests that drying may act as a relatively mild disturbance for aquatic invertebrate communities of Neotropical IRs.

While differences in the community composition among rivers with contrasting flow regimes are a consistent pattern in many climate regions worldwide, it is not clear whether such differences among flow regimes are consistent during and after periods of prolonged drought (*sensu* Lake, 2003) and how much these differences are affected by organismal dispersal modes and strengths (e.g. Leigh & Datry, 2016). Using biomonitoring data on aquatic invertebrate and fish communities in subtropical rivers in Australia, Rolls *et al.* (2016) tested whether beta diversity across naturally near-perennial, mildly intermittent and highly intermittent streams were greater during a 3-year drought characterised by low rainfall than during a wetter post-drought phase. They also tested whether differences in beta diversity between phases would decrease with increasing dispersal strength of organisms (Cañedo-Argüelles *et al.*, 2015; Datry *et al.*, 2015). Except for locally dispersing fish, beta-diversity differed among flow regimes for all dispersal groups of invertebrates and fish. Within flow-regime groups, beta-diversity was higher during the

drought than post-drought for most invertebrate groups (but was not different for fish) and, for most dispersal groups, was driven predominantly by replacement of taxa among samples rather than nestedness (except for diadromous fish). Overall, the findings suggest that beta diversity was similar during both phases, possibly due to consistent shifts in average community composition during recovery from drought in all flow regime groups. These consistent shifts may reflect some community resistance to the effects of drying–wetting cycles (Leigh *et al.*, 2016b) but also resilience and the presence of perennial refuges (e.g. water holes) in many of the rivers studied.

Where to next? A roadmap for ecological research on intermittent rivers

Contemporary IR ecology is a multifaceted and maturing field of research at the interface between aquatic and terrestrial ecology. The research findings in this Special Issue contribute to freshwater and general ecology by testing and challenging concepts across a range of topics in highly dynamic systems. This issue is also timely because information on IRs is being sought by managers and policy makers faced with increasing flow intermittence in response to climate change and growing demands for fresh water. There is ongoing debate regarding the legal status of IRs in many countries despite the increasing recognition of their ubiquity, their influence on perennial waters, and human-induced shifts increasing their prevalence and prolonging droughts (Acuña *et al.*, 2014; Datry *et al.*, 2014b). From the insights provided by the studies presented in this Special Issue, we draw several conclusions and proposals to direct future research and management.

Drying imposes a series of strong local filters expressed as incremental physicochemical and biological constraints selecting for biota and governing ecosystem processes (Williams, 1996; Boulton, 2003). Much of what is known about IR ecology has been informed by this perspective (Leigh *et al.*, 2016a). Comparative studies have used this perspective to test predictions regarding biodiversity and ecosystem processes between perennial and IRs as well as across continua of intermittence (Datry *et al.*, 2016b; Rolls *et al.*, 2016; Stromberg & Merritt, 2016). Although this Special Issue defines 'IRs' as any channel that periodically ceases to flow and dries, we recognise that IRs really represent a continuum with varying durations, frequencies, timing and expansion/contraction rates of drying (Stanley, Fisher & Grimm, 1997; Costigan *et al.*, 2016). The study of these continua

can therefore provide a deeper understanding and predictive capability regarding how drying influences biodiversity and ecosystem function. Experimental approaches that manipulate different dimensions of intermittency (Vander Vorste *et al.*, 2016) and the use of models (Fisher *et al.*, 1998; Datry *et al.*, 2016a) will also provide focused mechanistic advances.

Drying is increasingly recognised as affecting flow continuity through time and flow connectivity through space (e.g. Larned *et al.*, 2011; Bogan, Boersma & Lytle, 2013; Jaeger *et al.*, 2014). This aligns well with mainstream ecology's recent use of meta-system perspectives (e.g. Loreau, Mouquet & Holt, 2003; Datry *et al.*, 2016a,b,c) and riverine ecologists' growing interest in connectivity at the network scale (e.g. Carrara *et al.*, 2012; Zimmermann *et al.*, 2014). A common thread across most the papers in this Special Issue is the role of hydrological (dis)connectivity in IR ecology, whether by affecting dispersal or material flows (e.g. Whitney *et al.*, 2016). Connectivity underlies processes across all scales, from local within-habitat (Vander Vorste *et al.*, 2016) to longitudinal segments (Marshall *et al.*, 2016) to entire networks (Datry *et al.*, 2016a; Rolls *et al.*, 2016; Welter & Fisher, 2016). Longitudinal connectivity along IRs may receive the most attention because surface water is the dominant transport route for many riverine organisms and materials, however because of recurrent wet–dry cycles, the vertical connectivity with ground water and atmosphere can be important (Siebers *et al.*, 2016; Vander Vorste *et al.*, 2016; Welter & Fisher, 2016). In spite of the role surface–groundwater interactions play in shaping drying patterns in IRs, disruption of vertical connectivity has attracted little interest from ecologists (Leigh *et al.*, 2016a), reflecting the lag in recognition of the functional significance of hyporheic zone in perennial systems (Boulton *et al.*, 2010). Although rivers are linear features, lateral connectivity is critical in understanding resource subsidies (McCluney & Sabo, 2012) and community dynamics for floodplain systems (Fernandes *et al.*, 2014). Less studied has been the lateral movement by organisms that inhabit or use dry stream beds, and more research is needed to inform understanding of their persistence in IRs (Corti & Datry, 2016). In addition to movement across space, storage or the lack of movement can be critical in understanding the distribution and abundance of materials and organisms in IRs. Numerous species inhabiting IRs have resting stages and adaptations that enable them to withstand (resist) the drying conditions *in situ* and resume growth and reproduction after rewetting (Stromberg & Merritt, 2016; Stubbington *et al.*, 2016). The importance of resistance or storage to

the persistence or dynamics of populations and communities in IRs relative to other ecosystems, such as temporary pools, estuaries, or forest soils, is unknown. The papers presented here will encourage modification of existing approaches and development of novel approaches for investigating the (dis)continuity and (dis)connectivity in IRs and the degree to which these factors, as well as environmental filtering, govern biodiversity and biogeochemical patterns across ecosystems.

The study of IRs is situated at the interface of aquatic and terrestrial ecology. Most papers in this Special Issue emphasise the wet rather than the dry phase, however there is growing interest in dry phases and the terrestrial vegetation and invertebrate assemblages living in and along IR channels. Biogeochemistry and microbial studies are also areas where the dry phase is being incorporated into a more complete understanding of IR ecosystems. Interdisciplinary studies on IRs should be encouraged to further blur the demarcation between aquatic and terrestrial ecology (Datry *et al.*, 2014a). IRs are also extreme examples of systems where ecological patterns and processes are shaped by temporal environmental fluctuation. Therefore, IRs can be considered useful models for understanding seasonal food-web shifts over time (McMeans *et al.*, 2015) and where/how resource booms and busts interact and alternate among members of flowing and dry channel communities (Balcombe *et al.*, 2015). The compression of interactions and flows of materials and energy to particular points in time and space that may vary in their predictability is certainly an area where IR research can contribute to aquatic and general ecology (Datry *et al.*, 2016a).

As a result of ongoing climate change and human modification of landscapes and waterways, flooding and drought are expected to become more extreme and possibly more frequent (Ledger & Milner, 2015). Although flooding can be detrimental to society and riverine ecosystems, drying of formerly perennial rivers is expected to be more adverse and more difficult to mitigate through management actions (Palmer *et al.*, 2008). Shifts in streamflow regimes from perennial to intermittent and extension of the duration and frequency of dry phases in IRs could strongly affect river ecosystems and the services they provide (Costigan *et al.*, 2016; Datry *et al.*, 2016a,b,c). While this Special Issue testifies to recent advances in IR ecology, scientific aspects are certainly yet to be uncovered, and foreseeable challenges for their protection and restoration indicate there is still much room for growth and innovation.

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